ASHOK GAHATRAJ

Geospatial Researcher | Spatial Data Analytics | Remote Sensing

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RESEARCH PROFILE

Geospatial researcher specializing in GIS, remote sensing, and spatial analytics. Experienced in applying statistical modeling and machine learning to analyze environmental risk, urban systems, and public health. Skilled in Python, R, and the Esri ecosystem, with a strong interest in GeoAI and data-driven decision-making.

EDUCATION

Master's in Applied Geography — New Mexico State University, Las Cruces, NM *, Expected May 2026*
GPA: 3.97/4.0 | Thesis: *Remote Sensing of Vegetation Phenology in Southwestern U.S. Sky Islands: A Comparative Analysis of the Sonoran, Chihuahuan, and Mojave Deserts* | Advisor: Dr. Michaela Buenemann

B.E. in Geomatics Engineering — Tribhuvan University, Pokhara, Nepal *2017 – 2022*

SKILLS

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Online (Dashboards, Experience Builder, Hub), ArcGIS Indoors, QGIS, ENVI, Google Earth Engine, Geoprocessing
Programming & Data Science: Python, R, SQL, JavaScript | Spatial Analysis, Time-Series Analysis, Statistical Modeling, Data Pipelines
Machine Learning & GeoAI: Regression, Classification (RF, SVM), Cross-Validation, Spatial Modeling, Computer Vision (basic)
Data Visualization: Matplotlib, ggplot2, Power BI
Design & CAD: AutoCAD, Revit (basic)
Surveying & UAV: DGPS, Total Station, UAV Mapping & Photogrammetry
Languages: English (fluent), Nepali (native), Spanish (basic)

WORK EXPERIENCE

GIS Analyst Intern | New Mexico State University, Las Cruces, NM *May 2025 – Sept. 2025*
- Led design and deployment of a campus-wide ArcGIS Indoors Information Model — architected the data schema and integrated complex floor plans for research space management used by university leadership.
- Built dynamic dashboards in ArcGIS Online tracking occupancy, funding, and space utilization.
- Developed a campus ArcGIS Hub site combining indoor maps, wayfinding tools, and analytics for digital twin visualization.

Graduate Research and Teaching Assistant | New Mexico State University, Las Cruces, NM *Aug. 2024 – Present*
- Developed a MODIS/Landsat-based phenology analysis pipeline (R + GEE) to quantify climate-driven vegetation dynamics across desert ecosystems.
- Processed multi-year time-series datasets and automated extraction workflows using Python, R, Phenofit, BFAST, and LandTrendr.
- Analyzed urban growth patterns, vegetation change (NDVI), and land surface temperature using multi-temporal remote sensing datasets.
- Applied spatial analysis to examine links between environmental exposure and public health risk.
- Supported instruction in physical geography, GIS programming, and remote sensing through lab assistance and grading.

Data Analyst | Numeric Mind, Kathmandu, Nepal *Oct. 2022 – Jun. 2024*
- Conducted geospatial data analysis on a healthcare system project, building a predictive model using machine learning to identify spatial patterns and support data-driven decision-making.
- Automated data cleaning and validation pipelines for client datasets using Python and SQL, improving processing efficiency.
- Collaborated with project managers to develop spatial and analytical tools for business intelligence needs.

KEY PROJECTS

<u>Indoor Space Management System (ArcGIS Indoors):</u>

- Designed and implemented an ArcGIS Indoors geodatabase and floor-aware mapping system for campus facilities
- Developed interactive dashboards to monitor occupancy, research activity, and space utilization
- Built ArcGIS Hub site integrating indoor navigation, analytics, and digital twin visualization

<u>Mapping Heat Risk & Healthcare Inequity in New Mexico:</u>

- Identified statistically significant clusters of extreme heat exposure and social vulnerability using spatial statistics (e.g., hotspot analysis, clustering techniques)
- Quantified spatial disparities in healthcare access by analyzing proximity to Medicare/Medicaid facilities in high-risk zones
- Modeled relationships between socioeconomic factors (age, poverty, vulnerability indices) and heat exposure using regression and spatial analysis
- Integrated climate, demographic, and infrastructure datasets to map high-risk populations and support data-driven public health planning

<u>Urban Heat, Vegetation, and Park Accessibility in El Paso, TX:</u>

- Analyzed spatial relationships between vegetation (NDVI) and land surface temperature (LST) using spatial statistics (Moran's I, GWR) to identify urban heat patterns
- Evaluated spatial clustering of heat exposure and vegetation cover to detect hotspots and environmental disparities across neighborhoods
- Assessed accessibility to green spaces and parks to identify underserved areas and inform urban planning and heat mitigation strategies

CERTIFICATIONS

- Introduction to GIS Programming, UT Knoxville (2025)
- ArcGIS Indoors: Deploying Indoor Viewer and Mobile Apps (2025)
- Regression Analysis: Building a Regression Model Using ArcGIS Pro (2025)
- Google Data Analytics Specialization, Coursera

AWARDS & HONORS

- 3rd Place, SWAAG Conference Poster Award — Indoor Space Management Project (2026)
- Robert & Elizabeth Czerniak Endowed Scholarship (2025)
- Richard D. Wright Graduate Award for Excellence in Applied GIS (2025)

PUBLICATIONS & CONFERENCES

- **Gahatraj, A., Bhusal, K. (2025).** *Assessing the Relationship Between Land Use/Land Cover Change, NDVI, NDBI, and Land Surface Temperature: A 20-Year Analysis of Rupandehi District, Nepal.* doi: 10.4236/oalib.1114123
- **Conference Posters:** AAG Annual Conference 2025; SWAAG 2024; Esri User Conference 2025 (Attendee); Geomatics Engineering Association of Nepal 2021

PROFESSIONAL AFFILIATIONS

- American Association of Geographers (AAG)
- Gamma Theta Upsilon Honor Society
- Graduate GES Organization – Treasurer
- Nepal Engineering Council
- Nepalese Student Association (NeSA)